

72. Physical Medium Dependent sublayer and baseband medium, type 10GBASE-KR

72.1 Overview

This clause specifies the 10GBASE-KR PMD and the baseband medium. When forming a complete PHY, a PMD shall be connected to the appropriate sublayers (see Table 72–1), and with the management functions that are optionally accessible through the management interface defined in Clause 45, or equivalent.

Table 72–1—Physical Layer clauses associated with the 10GBASE-KR PMD

Associated clause	10GBASE-KR
46—XGMII ^a	Optional
47—XGXS and XAUI	Optional
49—10GBASE-R PCS	Required
51—10-Gigabit Serial PMA	Required
73—Auto-Negotiation for Backplane Ethernet	Required
74—BASE-R FEC	Optional
78—Energy-Efficient Ethernet	Optional

^aThe XGMII is an optional interface. However, if the XGMII is not implemented, a conforming implementation behaves functionally as though the RS and XGMII were present.

A 10GBASE-KR PHY with the optional Energy-Efficient Ethernet (EEE) capability may optionally enter the Low Power Idle (LPI) mode to conserve energy during periods of low link utilization. The “Assert LPI” request at the XGMII is encoded in the transmitted symbols. Detection of LPI signaling in the received symbols is indicated as “Assert LPI” at the XGMII. Upon the detection of “Assert LPI” at the XGMII, an energy-efficient 10GBASE-KR PHY continues transmitting for a predefined period, then ceases transmission and deactivates transmit functions to conserve energy. The PHY periodically transmits during this quiet period to allow the remote PHY to refresh its receiver state (e.g., timing recovery, adaptive filter coefficients) and thereby track long-term variations in the timing of the link or the underlying channel characteristics. If, during the quiet or refresh periods, normal interframes resume at the XGMII, the PHY reactivates transmit functions and initiates transmission. This transmission will be detected by the remote PHY, causing it to also exit the LPI mode.

72.2 Physical Medium Dependent (PMD) service interface

The 10GBASE-KR PMD utilizes the PMD service interface defined in 52.1.1. The PMD service interface is summarized as follows:

- a) PMD_UNITDATA.request (as defined in 52.1.1)
- b) PMD_UNITDATA.indication (as defined in 52.1.1)
- c) PMD_SIGNAL.indication

If EEE is supported, the following primitives are also defined on the PMD Service Interface:

```
PMD_RX_MODE.request(rx_mode)
PMD_TX_MODE.request(tx_mode)
```

These messages affect the PCS variables as described in 49.2.13.2.2.

72.2.1 PMD_RX_MODE.request

This primitive is generated by the PCS Receive Process when EEE is supported to indicate that the input signal is quiet and the PMA and PMD receiver may go into a low power mode. When EEE is not supported, the primitive is never invoked and the PMD behaves as if rx_mode = DATA.

72.2.1.1 Semantics of the service primitive

```
PMD_RX_MODE.request (rx_mode)
```

The rx_mode parameter takes on one of two values: QUIET or DATA.

72.2.1.2 When generated

The PCS generates this primitive to request the appropriate PMD receive LPI state.

72.2.1.3 Effect of receipt

When rx_mode is QUIET, the PMD receive function may deactivate functional blocks to conserve energy. When rx_mode is DATA, the PMD receive function operates normally.

72.2.2 PMD_TX_MODE.request

This primitive is generated by the PCS Transmit Process when EEE is supported to indicate that the PMA and PMD transmit functions may go into a low power mode and to disable the PMD transmitter. See subclause 72.6.5. When EEE is not supported, the primitive is never invoked and the PMD behaves as if tx_mode = DATA.

72.2.2.1 Semantics of the service primitive

```
PMD_TX_MODE.request (tx_mode)
```

The tx_mode parameter takes on one of three values: QUIET, ALERT, or DATA.

72.2.2.2 When generated

The PCS generates this primitive to request appropriate PMD transmit LPI state.

72.2.2.3 Effect of receipt

When tx_mode is QUIET, the PMD Transmit function may deactivate functional blocks to conserve energy. When tx_mode is ALERT, the PMD Transmit function transmits the alert pattern. And when it is DATA, the PMD Transmit function operates normally.

72.3 PCS requirements for Auto-Negotiation (AN) service interface

The PCS associated with this PMD shall support the AN service interface primitive AN_LINK.indication defined in 73.9. (See 49.2.16.)

72.4 Delay constraints

Predictable operation of the MAC Control PAUSE operation (Clause 31, Annex 31B) demands that there be an upper bound on the propagation delays through the network. This implies that MAC, MAC Control sublayer, and PHY implementers consider the delay maxima, and that network planners and administrators consider the delay constraints regarding concatenation of devices. A description of overall system delay constraints and the definitions for bit-times and pause_quanta can be found in 69.3.

The sum of the transmit and the receive delays contributed by the 10GBASE-KR PMD and medium shall be no more than 1024 bit times. It is assumed that the round-trip delay through the medium is 160 bit times.

72.5 PMD MDIO function mapping

The optional MDIO capability described in Clause 45 defines several variables that provide control and status information for and about the PMD. If MDIO is implemented, it shall map MDIO control variables to PMD control variables as shown in Table 72–2, and MDIO status variables to PMD status variables as shown in Table 72–3.

Table 72–2—MDIO/PMD control variable mapping

MDIO control variable	PMA/PMD register name	Register/ bit number	PMD control variable
Reset	Control register 1	1.0.15	PMD_reset
Global PMD Transmit Disable	Transmit disable register	1.9.0	Global_PMD_transmit_disable
Restart training	BASE-R PMD control register	1.150.0	mr_restart_training
Training enable	BASE-R PMD control register	1.150.1	mr_training_enable

72.6 PMD functional specifications

72.6.1 Link block diagram

For purposes of system conformance, the PMD sublayer is standardized at test points TP1 and TP4 as shown in Figure 72–1. The transmitter and receiver blocks include all off-chip components associated with the respective block. For example, external AC-coupling capacitors, if required, are to be included in the receiver block.

The electrical path from the transmitter block to TP1, and from TP4 to the receiver block, will affect link performance and the measured values of electrical parameters used to verify conformance to this standard. Therefore, it is recommended that this path be carefully designed.

Table 72–3—MDIO/PMD status variable mapping

MDIO status variable	PMA/PMD register name	Register/ bit number	PMD status variable
Fault	Status register 1	1.1.7	PMD_fault
Transmit fault	Status register 2	1.8.11	PMD_transmit_fault
Receive fault	Status register 2	1.8.10	PMD_receive_fault
Global PMD Receive signal detect	Receive signal detect register	1.10.0	Global_PMD_signal_detect
Receiver status	BASE-R PMD status register	1.151.0	rx_trained
Frame lock	BASE-R PMD status register	1.151.1	frame_lock
Startup protocol status	BASE-R PMD status register	1.151.2	training
Training failure	BASE-R PMD status register	1.151.3	training_failure

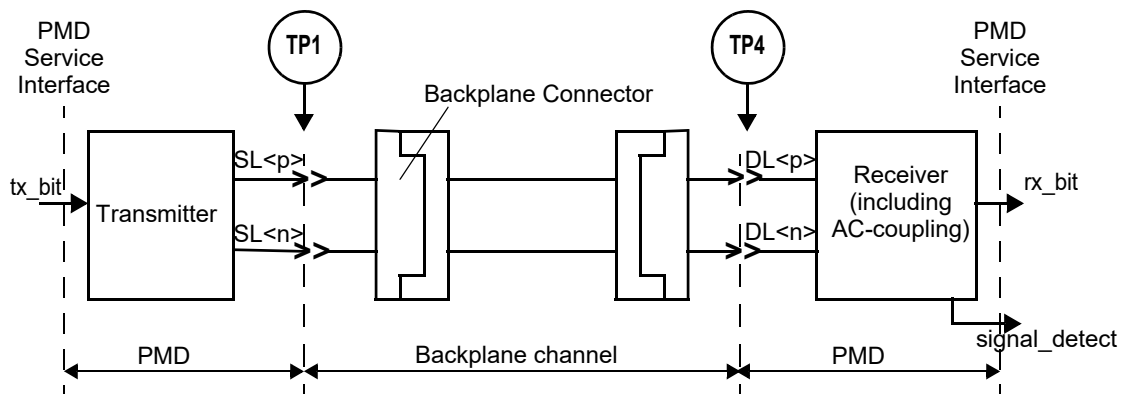


Figure 72–1—Link block diagram

72.6.2 PMD transmit function

The PMD Transmit function shall convey the bits requested by the PMD service interface message `PMD_UNITDATA.request(tx_bit)` to the MDI according to the specifications in this clause. A positive output voltage of $SL<p>$ minus $SL<n>$ (differential voltage) shall correspond to $tx_bit = ONE$.

If the optional Energy-Efficient Ethernet (EEE) capability is supported (see Clause 78) then when tx_mode is set to ALERT, the PMD will transmit a repeating 16-bit pattern, hexadecimal 0xFF00. When tx_mode is ALERT, the transmitter equalizer taps are set to the preset state specified in 72.6.10.2.3.1. When tx_mode is DATA, the driver coefficients are restored to their states resolved during training.

72.6.3 PMD receive function

The PMD Receive function shall convey the bits received from the MDI according to the electrical specifications in this clause to the PMD service interface using the message `PMD_UNITDATA.indication(rx_bit)`. A positive input voltage of $DL<p>$ minus $DL<n>$ (differential voltage) shall correspond to $rx_bit = ONE$.

72.6.4 PMD signal detect function

The Global PMD signal detect function shall report to the PMD service interface, using the message `PMD_SIGNAL.indication(SIGNAL_DETECT)`, which is signaled continuously. `PMD_SIGNAL.indication` is used by 10GBASE-KR to indicate the successful completion of the startup protocol. When the PHY supports the optional EEE capability, `PMD_SIGNAL.indication` is also used to indicate when the ALERT signal is detected, which corresponds to the beginning of a refresh or a wake. If the MDIO interface is implemented, then `Global_PMD_signal_detect (1.10.0)` shall be continuously set to the value of `SIGNAL_DETECT` as described in 45.2.1.9.7.

The value of the `SIGNAL_DETECT` is defined by the training state diagram shown in Figure 72–5 when the PHY does not support the EEE capability or if the PHY supports the EEE capability and `rx_mode` is set to DATA. When the PHY supports the EEE capability, `SIGNAL_DETECT` is set to FAIL following a transition from `rx_mode = DATA` to `rx_mode = QUIET`. When `rx_mode = QUIET`, `SIGNAL_DETECT` shall be set to OK within 500 ns following the application of a signal at the receiver input that is the output of a channel that satisfies the requirements of all the parameters of both interference tolerance test channels defined in 72.7.2.1 when driven by a square wave pattern with a period of 16 unit intervals and peak-to-peak differential output amplitude of 720 mV. While `rx_mode = QUIET`, `SIGNAL_DETECT` changes from FAIL to OK only after a valid ALERT signal is applied to the channel.

`SIGNAL_DETECT` shall be set to FAIL following system reset or the manual reset of the training state diagram. Upon completion of training, `SIGNAL_DETECT` shall be set to OK.

If training is disabled by management and EEE is not implemented, `SIGNAL_DETECT` shall be set to OK.

72.6.5 PMD transmit disable function

The `Global_PMD_transmit_disable` function is mandatory if EEE is supported and is otherwise optional. When this function is supported, it shall meet the requirements of this subclause.

- a) When the `Global_PMD_transmit_disable` variable is set to ONE, this function shall turn off the transmitter such that it drives a constant level (i.e., no transitions) and does not exceed the maximum differential peak-to-peak output voltage specified in Table 72–6.
- b) If a `PMD_fault (72.6.7)` is detected, then the PMD may turn off the electrical transmitter.
- c) Loopback, as defined in 72.6.6, shall not be affected by `Global_PMD_transmit_disable`.
- d) For EEE capability, the `PMD_transmit_disable` function shall turn off the transmitter after `tx_mode` is set to QUIET within a time and voltage level specified in 72.7.1.4. The `PMD_transmit_disable` function shall turn on the transmitter after `tx_mode` is set to DATA or ALERT within the time and voltage level specified in 72.7.1.4.

If the MDIO interface is implemented, then this function shall map to the `Global_PMD_transmit_disable` bit as specified in 45.2.1.8.7.

72.6.6 Loopback mode

Loopback mode shall be provided for the 10GBASE-KR PMD by the transmitter and receiver of a device as a test function to the device. When loopback mode is selected, transmission requests passed to the transmitter are shunted directly to the receiver, overriding any signal detected by the receiver on its attached link. Note, this bit does not affect the state of the transmitter. The method of implementing loopback mode is not defined by this standard.

Control of the loopback function is specified in 45.2.1.1.5.

NOTE 1—The signal path that is exercised in the loopback mode is implementation specific, but it is recommended that this signal path encompass as much of the circuitry as is practical. The intention of providing this loopback mode of operation is to permit diagnostic or self-test functions to test the transmit and receive data paths using actual data. Other loopback signal paths may also be enabled independently using loopback controls within other devices or sublayers.

NOTE 2—Placing a network port into loopback mode can be disruptive to a network.

72.6.7 PMD_fault function

If the MDIO is implemented, PMD_fault is the logical OR of PMD_receive_fault, PMD_transmit_fault, and any other implementation specific fault.

72.6.8 PMD transmit fault function

The PMD_transmit_fault function is optional. The faults detected by this function are implementation specific, but should not include the assertion of the Global_PMD_transmit_disable function.

If a PMD_transmit_fault (optional) is detected, then the Global_PMD_transmit_disable function should also be asserted.

If the MDIO interface is implemented, then this function shall be mapped to the PMD_transmit_fault bit as specified in 45.2.1.7.4.

72.6.9 PMD receive fault function

The PMD_receive_fault function is optional. The faults detected by this function are implementation specific.

If the MDIO interface is implemented, then this function shall contribute to PMA/PMD receive fault bit as specified in 45.2.1.7.5.

72.6.10 PMD control function

72.6.10.1 Overview

The PMD control function generates the control actions required to bring the PMD from initialization to a mode in which data may be exchanged with the link partner.

The PMD control function implements the 10GBASE-KR startup protocol. This protocol facilitates timing recovery and equalization while also providing a mechanism through which the receiver can tune the transmit equalizer to optimize performance over the backplane interconnect. The protocol supports these mechanisms through the continuous exchange of fixed-length training frames.

If IEEE 802.3-2022 is supported, the PMD control function responds to PCS requests to transition in and out of quiet states.

72.6.10.2 Training frame structure

The training frame is a fixed length structure that is sent continuously during training. The training frame, shown in Figure 72–2, is 548 octets in length and contains a control channel and training pattern.

The control channel is signaled using differential Manchester encoding (DME) at a signaling rate equal to one quarter of the 10GBASE-KR signaling rate. Since each DME symbol contains two DME transition positions and each transition position is four 10GBASE-KR UI, one control channel bit is transmitted every eight 10GBASE-KR UI.

Differential Manchester encoding guarantees transition density and DC balance while the reduced rate of transmission facilitates reception over non-optimally equalized channels.¹³⁴

Training frames are delimited by a fixed 4 octet frame marker.

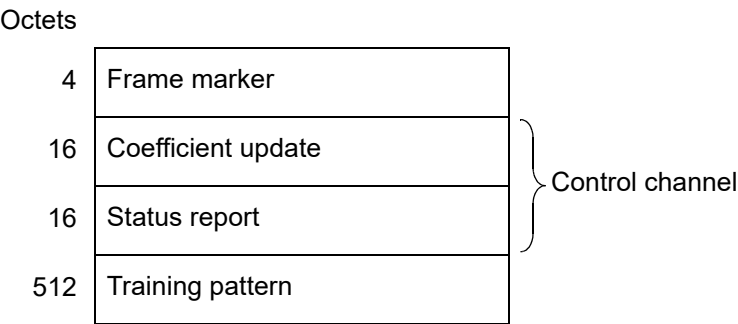


Figure 72–2—Training frame structure

72.6.10.2.1 Frame marker

Frames are delimited by the 32-bit pattern, hexadecimal 0xFFFF0000 (ones transmitted first), as expressed in 10.3125 Gbd symbols. This pattern does not appear in the control channel or the training pattern and therefore serves as a unique indicator of the start of a training frame.

72.6.10.2.2 Control channel encoding

The control channel shall be transmitted using differential Manchester encoding (DME). The rules of differential Manchester encoding are as follows:

- a) A data transition shall occur at each cell boundary.
- b) A mid-cell data transition shall be used to signal a logical one.
- c) The absence of a mid-cell data transition shall be used to signal a logical zero.

If a coding violation is detected within the bounds of the control channel in a given training frame, the contents of the control channel for that frame shall be ignored.

The data cell length shall be 8 10GBASE-KR UI. Therefore, the total length of the control channel is 256 10GBASE-KR UI.

72.6.10.2.3 Coefficient update field

The coefficient update field carries correction information from the local receiver to the link partner transmit equalizer. The field consists of preset controls, initialization controls, and coefficient updates for three transmit equalizer taps. The format of the coefficient update field shall be as shown in Table 72–4. Cell 15 of the coefficient update field sent shall be transmitted first. The preset, initialize, and coefficient updates are set by the receiver adaptation process. The algorithm employed by the receiver adaptation process is beyond the scope of this standard.

¹³⁴The differential Manchester encoding defined for Backplane Ethernet is different from that defined in IEEE Std 802.5™.

Table 72–4—Coefficient update field

Cell(s)	Name	Description
15:14	Reserved	Transmitted as 0, ignored on reception.
13	Preset	1 = Preset coefficients 0 = Normal operation
12	Initialize	1 = Initialize coefficients 0 = Normal operation
11:6	Reserved	Transmitted as 0, ignored on reception.
5:4	Coefficient (+1) update	5 4 1 1 = reserved 0 1 = increment 1 0 = decrement 0 0 = hold
3:2	Coefficient (0) update	3 2 1 1 = reserved 0 1 = increment 1 0 = decrement 0 0 = hold
1:0	Coefficient (–1) update	1 0 1 1 = reserved 0 1 = increment 1 0 = decrement 0 0 = hold

72.6.10.2.3.1 Preset

The preset control is sent to request that the coefficients be set to a state where equalization is turned off. When received, the pre-cursor ($k = -1$) and post-cursor ($k = +1$) coefficients shall be set to a zero value and the main ($k = 0$) coefficient shall be set to its maximum value. The preset control shall only be initially sent when all coefficient status fields indicate not_updated, and will then continue to be sent until the status for all coefficients indicates updated or maximum. At that point, the outgoing preset control shall be set to zero. Maximum status shall be returned when the main coefficient is updated. Maximum status shall be returned for the pre-cursor and/or post-cursor coefficients when the coefficient is updated and zero is its maximum supported value. Updated status shall be returned for the pre-cursor and/or post-cursor coefficients when the coefficient is updated and it supports additional settings above the value zero.

A new request to preset or initialize shall not be sent until the incoming status messages for all coefficients revert to not_updated. Preset shall not be sent in combination with initialize or coefficient increment/decrement requests.

72.6.10.2.3.2 Initialize

The initialize control is sent to request that the coefficients be set to configure the transmit equalizer to its INITIALIZE state. When received, the taps shall be set such that the transmit output meets the conditions defined in 72.6.10.4.2. The initialize control shall only be initially sent when all coefficient status fields indicate not_updated, and will then continue to be sent until no coefficient status field indicates not_updated.

Updated status shall be returned for each coefficient when the coefficient update is completed. At that point, the outgoing initialize control shall be set to zero.

A new request to preset or initialize shall not be sent until the incoming status messages for all coefficients revert to not_updated. Initialize shall not be sent in combination with coefficient increment/decrement requests or preset.

72.6.10.2.3.3 Coefficient (k) update

Each coefficient, k , is assigned a 2-bit field describing a requested update. Three request encodings are defined: increment, decrement, and hold. The default state for a given tap is hold, which corresponds to no change in the coefficient. The increment or decrement encodings are transmitted to request that the corresponding coefficient be increased or decreased. The amount of change implemented by the transmitter in response to the coefficient update request shall meet the requirements of Table 72–7 and 72.7.1.10. An increment or decrement request shall continue to be transmitted until the update status for that tap (as defined in 72.6.10.2.4.5) indicates updated, maximum, or minimum. At that point, the outgoing requests for that tap shall be set to hold.

A new request to increment or decrement shall not be sent before the incoming status messages for that tap revert to not_updated. Coefficient increment/decrement shall not be sent in combination with initialize or preset.

The valid range for k is -1 to $+1$ where $k = 0$ denotes the main tap. The encoding of the coefficient update shall be as shown in Table 72–4.

72.6.10.2.4 Status report field

The status report field is used to signal state information from the local PMD to the link partner. The format of the status report field shall be as shown in Table 72–5. Cell 15 of the status report field shall be transmitted first.

Table 72–5—Status report field

Cell(s)	Name	Description
15	Receiver ready	1 = The local receiver has determined that training is complete and is prepared to receive data. 0 = The local receiver is requesting that training continue.
14:6	Reserved	Transmitted as 0, ignored on reception.
5:4	Coefficient (+1) status	5 4 1 1 = maximum 1 0 = minimum 0 1 = updated 0 0 = not_updated
3:2	Coefficient (0) status	3 2 1 1 = maximum 1 0 = minimum 0 1 = updated 0 0 = not_updated
1:0	Coefficient (–1) status	1 0 1 1 = maximum 1 0 = minimum 0 1 = updated 0 0 = not_updated

72.6.10.2.4.4 Receiver ready

The receiver ready bit is used to signal the local receiver state to the link partner. When asserted, the receiver ready bit indicates that the local receiver has concluded training and is prepared to receive data. When de-asserted, the receiver ready bit indicates that the local receiver is requesting that training continue. The format of the receiver ready bit shall be as shown in Table 72–5.

72.6.10.2.4.5 Coefficient (*k*) status

Each coefficient, *k*, is assigned a 2-bit field describing the status of pending updates to the coefficient. Four status encodings are defined: not updated, updated, maximum, and minimum.

These status encodings indicate the corresponding state of the coefficient update state diagram for coefficient *k*.

The valid range for *k* is –1 to +1 where *k* = 0 denotes the main tap. The encoding of the coefficient update shall be as shown in Table 72–5.

72.6.10.2.5 Coefficient update process

Each coefficient, k , has an associated coefficient update state diagram that controls updates of the coefficient and generates the tap update status field.

The default state for a given tap is `not_updated`. An increment or decrement request will only be acted upon when the state of the tap is `not_updated`. Upon execution of a received increment or decrement request, the status is reported as `updated`, `maximum`, or `minimum`. `Maximum` is reported if a received increment request causes the tap value to reach its maximum limit, or if it is already at that limit. `Minimum` is reported if a received decrement request causes the tap value to reach its minimum limit, or if it is already at that limit.

Once the `updated`, `maximum`, or `minimum` state is reported it continues to be reported until a `hold` request is received, after which the status reverts to `not_updated`.

The coefficient update process responds to coefficient requests as specified in the state diagram shown in Figure 72–6.

72.6.10.2.6 Training pattern

The training pattern shall be a 512 octet pattern consisting of 4094 bits from the output of a pseudo-random bit sequence of order 11 (PRBS11) generator followed by two zeros. The PRBS11 pattern generator shall produce the same result as the implementation shown in Figure 72–3. This implements the generator polynomial shown in Equation (72–1).

$$G(x) = 1 + x^9 + x^{11} \quad (72-1)$$

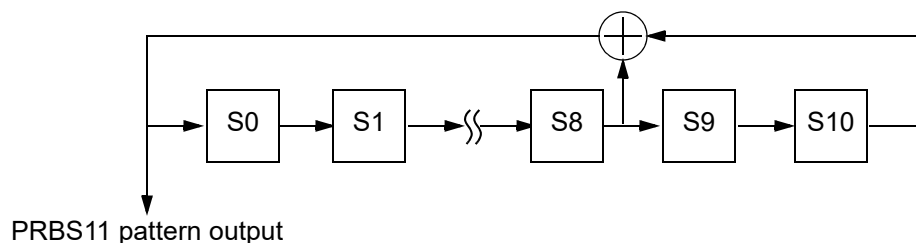


Figure 72–3—PRBS11 pattern generator

The pseudo-random generator shall have a random seed at the start of the training pattern. Each bit of the training pattern is transmitted as a single 10.3125 GBd symbol.

72.6.10.3 State variables

The notation used in the state diagrams follows the conventions of 21.5. State diagram timers follow the conventions of 14.2.3.2. The notation ++ after a counter or integer variable indicates that its value is to be incremented.

72.6.10.3.1 Variables

coefficient

Integer variable containing a value that should be used as the tap coefficient.

dec

Boolean variable that is set to TRUE when a training frame has been completely received and the coefficient update field of that frame for this coefficient is decrement, and is set to FALSE on reception of any other value.

frame_lock

Boolean variable that is set to TRUE when the receiver acquires training frame delineation and is set to FALSE otherwise.

frame_offset

Boolean variable that is set to TRUE after receiving one full training frame (548 octets) from the current frame start position. The Boolean variable is set to FALSE when the GET_NEW_MARKER state is entered. The current frame start position is indicated by a transition into the GET_NEW_MARKER state when the Boolean variable is set to FALSE.

hold

Boolean variable that is set to TRUE when a training frame has been completely received and the coefficient update field of that frame for this coefficient is hold, and neither preset or initialize are activated, and is set to FALSE on reception of any other value.

inc

Boolean variable that is set to TRUE when a training frame has been completely received and the coefficient update field of that frame for this coefficient is increment, and set to FALSE on reception of any other value.

initialize

Boolean variable that is set to TRUE when a training frame has been completely received and the initialize field of that frame is set to one and the preset field is set to zero, and is set to FALSE otherwise.

local_rx_ready

Boolean variable that is set to TRUE by the training state diagram when rx_trained is asserted and is set to FALSE otherwise. This value is transmitted as the receiver ready bit on all outgoing training frames.

marker_valid

Boolean variable that is set to TRUE when the candidate frame marker matches the specified frame marker pattern and is set to FALSE when the candidate frame marker does not match the specified frame marker pattern.

max_limit

Integer variable containing the maximum tap coefficient value, subject to the constraints detailed in 72.7.1.10.

min_limit

Integer variable containing the minimum tap coefficient value, subject to the constraints detailed in 72.7.1.10.

mr_restart_training

Boolean variable used by system management to restart the 10GBASE-KR startup protocol. When set to TRUE, it forces the training state diagram to the INITIALIZE state.

mr_training_enable

Boolean variable used by system management to enable or disable the 10GBASE-KR startup

protocol. It is set to TRUE when the startup protocol is enabled and set to FALSE when the startup protocol is disabled.

new_coeff

Integer variable containing the result of increment/decrement operations on the coefficient value

new_marker

Boolean variable that is set to TRUE when a new candidate frame marker is available for testing and FALSE when the TEST_MARKER state is entered. A new marker is available for testing when the training frame lock process has accumulated one frame marker (4 octets) from a candidate frame start position.

preset

Boolean variable that is set to TRUE when a training frame has been completely received and the preset field of that frame is set to one and is set to FALSE if set to zero.

remote_rx_ready

Boolean variable that is set to FALSE upon entry into the SEND_TRAINING state. The value of remote_rx_ready shall not be set to TRUE until no fewer than three consecutive training frames have been received with the receiver ready bit asserted.

reset

Boolean variable that controls the resetting of the PMA/PMD. It is set to TRUE whenever a reset is necessary including when reset is initiated from the MDIO, during power on, and when the MDIO has put the PMA/PMD into low-power mode.

rx_trained

Boolean variable that is set to TRUE when the remote transmit and local receive equalizers have been optimized and normal data transmission may commence and set to FALSE otherwise.

signal_detect

Boolean variable that is set to TRUE when the training process is complete and is set to FALSE otherwise. The value of signal_detect is reported to the PMA sublayer via the PMD_SIGNAL.indication primitive.

slip_done

Boolean variable that is set to TRUE when the SLIP requested by the Frame Lock State Diagram has been completed indicating that the next candidate frame sync position can be tested.

training

Boolean variable that is set to TRUE to indicate that the 10GBASE-KR startup protocol is in progress and is set to FALSE when training has completed.

training_failure

Boolean variable that is set to TRUE when the training state diagram has timed out due to expiration of the max_wait_timer while in the SEND_TRAINING, TRAIN_LOCAL, or TRAIN_REMOTE states and is set to FALSE otherwise.

update_status

Value to be transmitted in the Coefficient Status field for this coefficient in the next transmitted training frame, as defined in Table 72–8.

72.6.10.3.2 Timers

max_wait_timer

This timer is started in the INITIALIZE state of the training state diagram. If the max_wait_timer expires the training state diagram will enter the TRAINING_FAILURE state. The value of max_wait_timer shall be 500 ms $\pm$ 1%.

wait_timer

This timer is started when the local receiver is trained and detects that the remote receiver is ready to receive data. The local PMD will deliver wait_timer additional training frames to ensure that the link partner correctly detects the local receiver state. The value of wait_timer shall be between 100 and 300 training frames.

72.6.10.3.3 Counters

bad_markers

Count of the number of consecutive frame marker mis-matches.

good_markers

Count of the number of consecutive frame marker matches.

72.6.10.3.4 Functions

COEFF_UPDATE(coefficient, preset, initialize, inc, dec)

Returns an updated coefficient based on the contents of the coefficient update field in the training frame. Sets a fixed coefficient value, or adds, or subtracts from the current coefficient value to create the updated coefficient. If multiple actions are requested in the coefficient update field, then the priority is:

- 1) preset
- 2) initialize
- 3) inc/dec

Values: preset; If preset is TRUE then the function returns the coefficient value equivalent to no equalization [$c(-1)$ and $c(1)$ coefficients are set to zero, $c(0)$ set to maximum].
 initialize; If initialize is TRUE, then the function returns the coefficient value such that the transmit output meets the conditions defined in 72.6.10.4.2.
 inc; If inc is TRUE then the function returns (coefficient + step).
 dec; If dec is TRUE then the function returns (coefficient – step).

The requirements for the value of step are defined in 72.7.1.10 and Table 72–7.

SLIP

Causes the next candidate frame sync position to be tested. The precise method for determining the next candidate frame sync position is not specified and is implementation dependent. However, an implementation shall ensure that all possible bit positions are evaluated.

TRANSMIT(TRAINING, DATA)

Controls the output of the TRANSMIT functional block.

Values: TRAINING; the transmit block output is a continuous stream of training frames as defined in 72.6.10.2.
 DATA; the transmit block output is determined by the value of the input tx_bit.

72.6.10.4 State diagrams

72.6.10.4.1 Frame lock

The 10GBASE-KR PMD shall implement the Frame Lock state diagram as depicted in Figure 72–4 including compliance with the associated state variables as specified in 72.6.10.3. The frame lock state diagram determines when the PMD control function has detected the frame boundaries in the received data stream.

72.6.10.4.2 Training

The 10GBASE-KR PMD shall implement the Training state diagram as depicted in Figure 72–5 including compliance with the associated state variables as specified in 72.6.10.3. The training state diagram defines the operation of the 10GBASE-KR startup protocol. When the training state diagram enters the INITIALIZE state, the transmitter equalizer shall be configured such that R_{pre} and R_{pst} are $1.29 \pm 10\%$ and $2.57 \pm 10\%$ respectively. R_{pre} and R_{pst} are defined in 72.7.1.11. At the start of training the initial value of $c(0)$ shall be set such that the constraints of 72.7.1.11 are satisfied and the peak-to-peak differential output voltage shall be greater than or equal to 800 mV for a 1010 pattern.

72.6.10.4.3 Coefficient update

For each tap, the 10GBASE-KR PMD shall implement an instance of the coefficient update state diagram as depicted in Figure 72–6 including compliance with the associated state variables as specified in 72.6.10.3. The coefficient update state diagram defines the process for updating transmit equalizer coefficients in response to requests from the link partner, and also defines the coefficient update status to be reported in outgoing training frames.

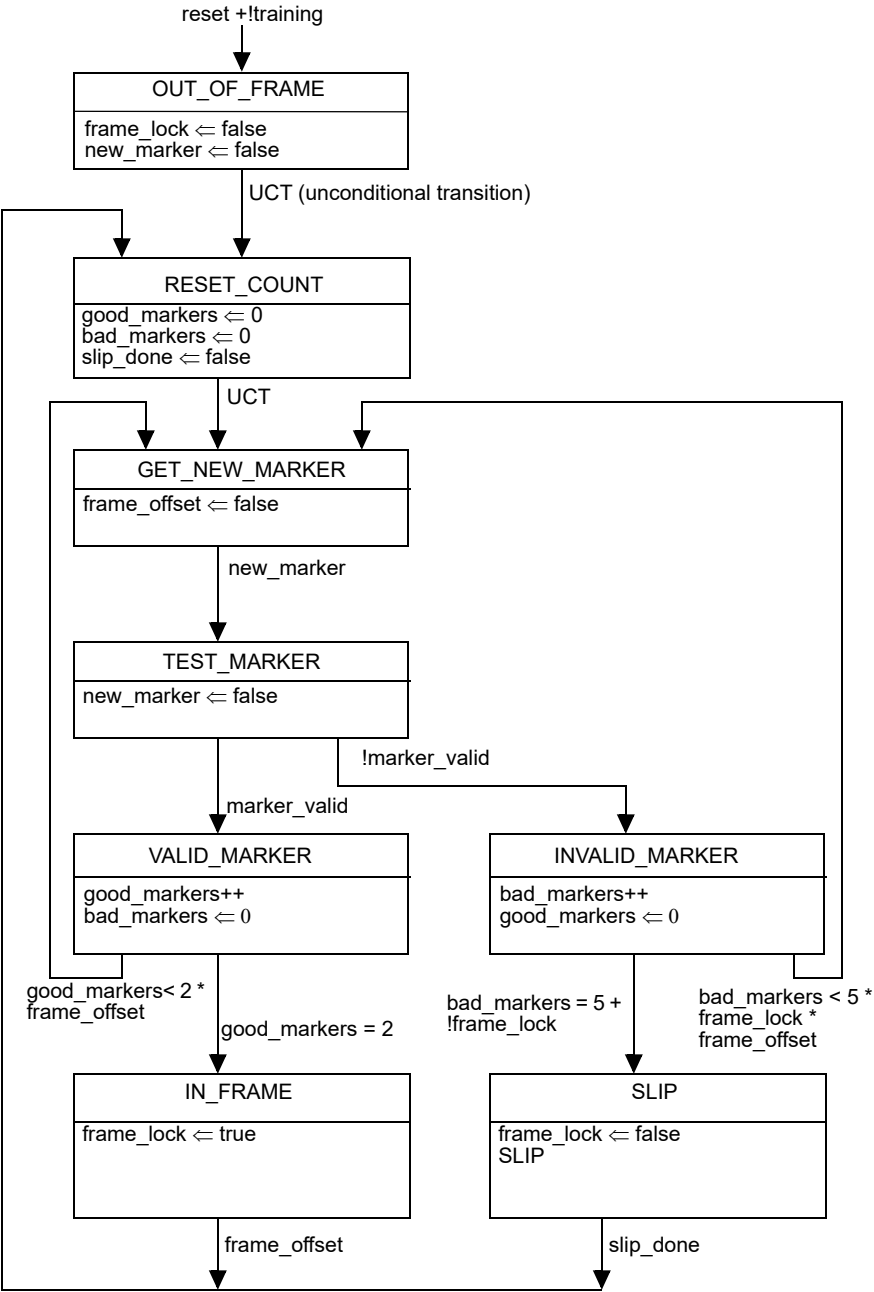


Figure 72-4—Frame lock state diagram

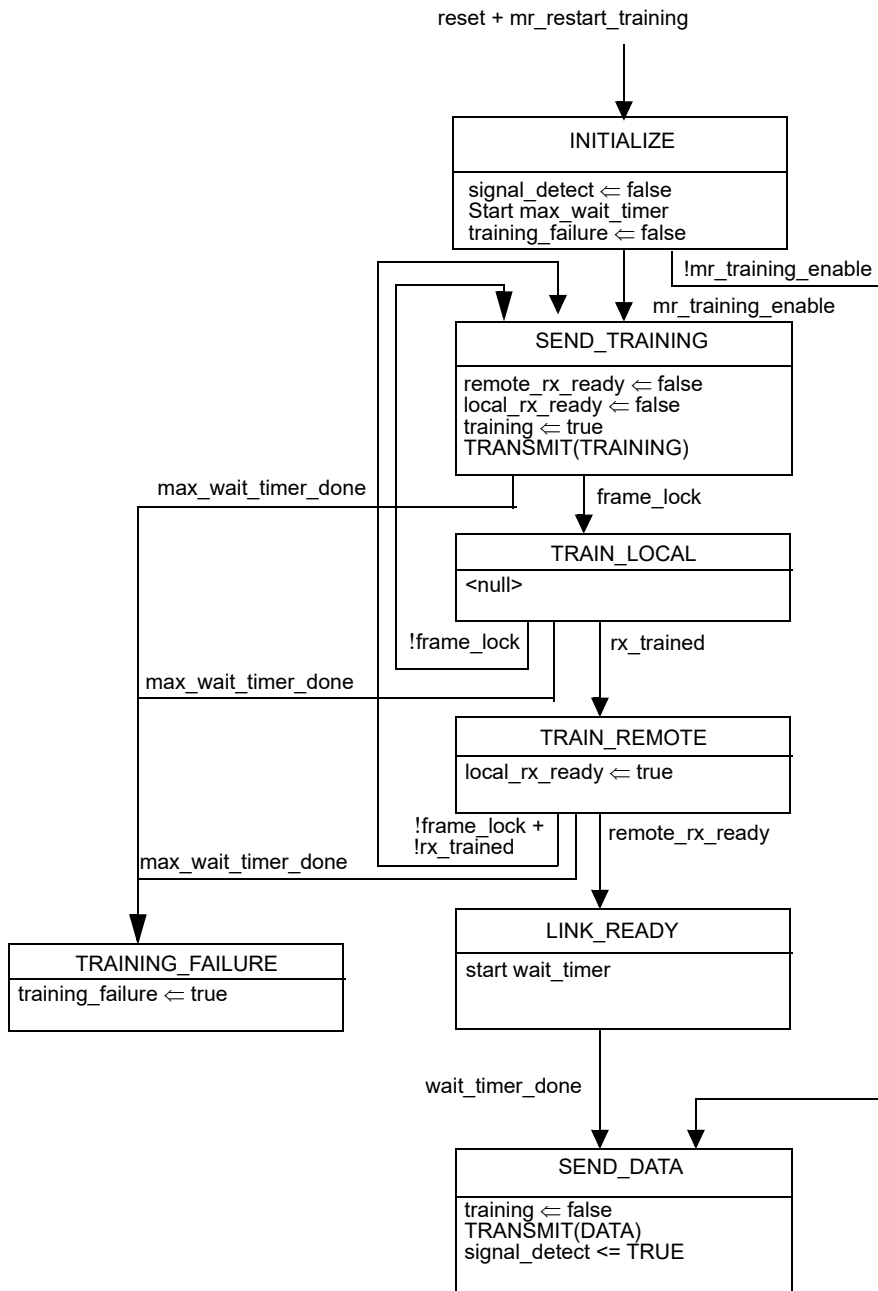


Figure 72–5—Training state diagram

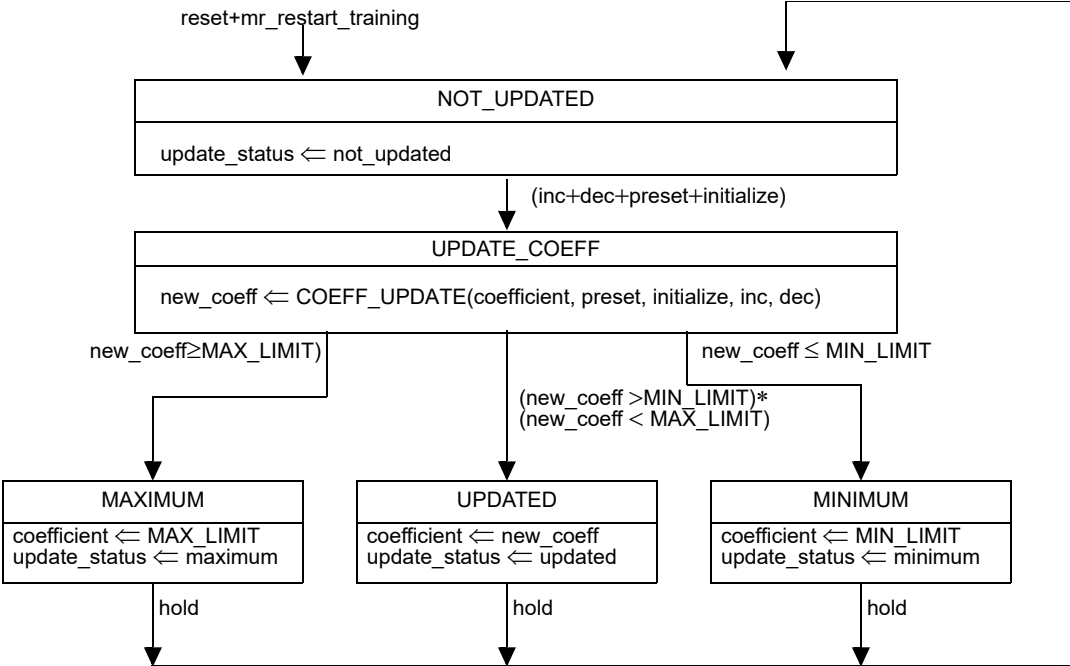


Figure 72-6—Coefficient update state diagram

72.6.11 PMD LPI function

The PMD LPI function responds to the transitions between Active, Sleep, Quiet, Refresh, and Wake states via the PMD_TX_MODE and PMD_RX_MODE requests. Implementation of the function is optional. EEE capabilities and parameters will be advertised during the Backplane Auto-negotiation, as described in 45.2.7.14. The transmitter on the local device will inform the link partner’s receiver when to sleep, refresh and wake. The local receiver transitions are controlled by the link partner’s transmitter and can change independent of the local transmitter states and transitions.

72.7 10GBASE-KR electrical characteristics

72.7.1 Transmitter characteristics

Transmitter characteristics at TP1 are summarized in Table 72–6 and detailed in 72.7.1.1 through 72.7.1.11.

Table 72–6—Transmitter characteristics for 10GBASE-KR

Parameter	Subclause reference	Value	Units
Signaling speed	72.7.1.3	10.3125 ± 100 ppm	GBd
Differential peak-to-peak output voltage (max.)	72.7.1.4	1200	mV
Differential peak-to-peak output voltage (max.) with TX disabled	72.6.5	30	mV
Common-mode voltage limits	72.7.1.4	0 to 1.9	V
Common-mode voltage deviation (max) during LPI	72.7.1.4	150	mV
Differential output return loss (min.)	72.7.1.5	[See Equation (72–4) and Equation (72–5)]	dB
Common-mode output return loss (min.)	72.7.1.6	[See Equation (72–6) and Equation (72–7)]	dB
Transition time (20%–80%)	72.7.1.7	24 to 47	ps
Max output jitter (peak-to-peak)	72.7.1.9		
Random jitter ^a		0.15	UI
Deterministic jitter		0.15	UI
Duty Cycle Distortion ^b		0.035	UI
Total jitter		0.28	UI

^aJitter is specified at BER 10^{-12} .

^bDuty Cycle Distortion is considered part of the deterministic jitter distribution.

72.7.1.1 Test fixture

The test fixture of Figure 72–7 or its functional equivalent, is required for measuring the transmitter specifications described in 72.7.1, with the exception of return loss.

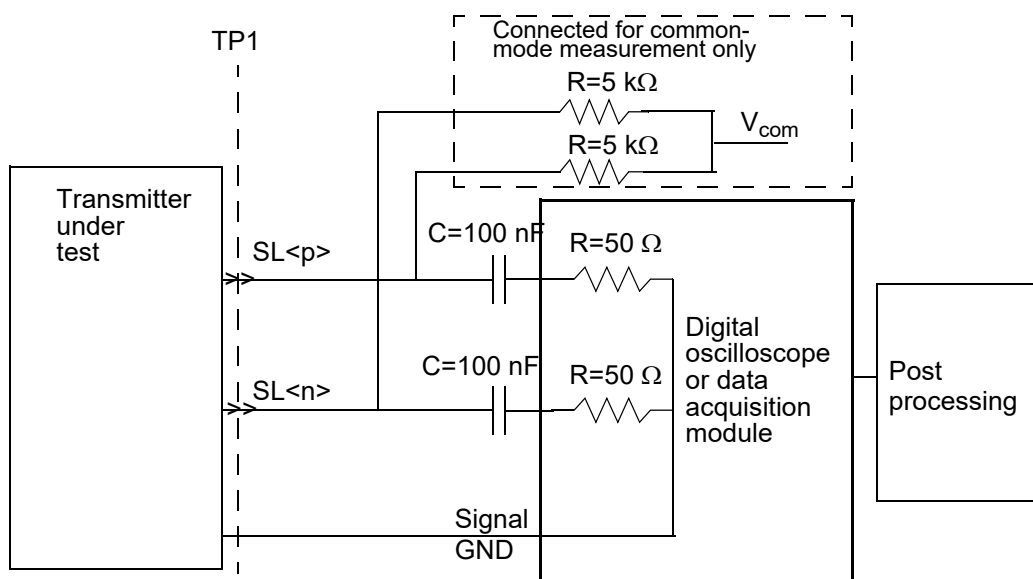


Figure 72-7—Transmit test fixture for 10GBASE-KR

72.7.1.2 Test fixture impedance

The differential load impedance applied to the transmitter output by the test fixture depicted in Figure 72-7 shall be 100 Ω. The differential return loss, in dB with f in MHz, of the test fixture shall meet the requirements of Equation (72-2) and Equation (72-3).

$$ReturnLoss(f) \geq 15 \quad (72-2)$$

for 100 MHz $\leq f < 5000$ MHz

$$ReturnLoss(f) \geq 15 - 26.57 \log_{10} \left(\frac{f}{5000 \text{ MHz}} \right) \quad (72-3)$$

for 5000 MHz $\leq f \leq 10000$ MHz

72.7.1.3 Signaling speed

The 10GBASE-KR signaling speed shall be 10.3125 GBd $\pm$ 100 ppm.

72.7.1.4 Output amplitude

The differential output voltage is constrained via the transmitter output waveform requirements specified in 72.7.1.10. For a 1010 pattern, the peak-to-peak differential output voltage shall be less than 1200 mV, regardless of equalization setting. The transmitter output voltage shall be less than 30 mV peak-to-peak when disabled. The differential output voltage test pattern shall consist of no fewer than eight symbols of alternating polarity.

NOTE 1—The required test patterns may be found in the training pattern field of the training frames or test patterns 2 or 3 as defined in 52.9.1.1.

NOTE 2—See Figure 72-8 for an illustration of the definition of differential peak-to-peak output voltage.

DC-referenced voltage levels are not defined since the receiver is AC-coupled. The common-mode voltage of SL<p> and SL<n> shall be between 0 V and 1.9 V with respect to signal ground as measured at V_{com} in Figure 72–7.

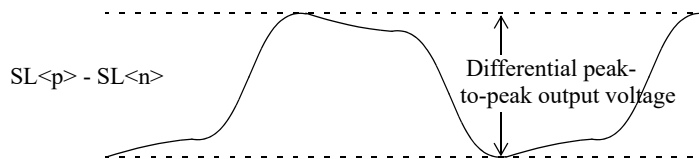


Figure 72–8—Transmitter differential peak-to-peak output voltage definition

NOTE—SL<p> and SL<n> are the positive and negative sides of the differential signal pair.

For EEE capability, the transmitter’s differential peak-to-peak output voltage shall be less than 30 mV within 500 ns of tx_mode being set to QUIET and remain so while tx_mode is set to QUIET. Furthermore, the transmitter’s differential peak-to-peak output voltage shall be greater than 720 mV within 500 ns of tx_mode being set to ALERT. The transmitter output shall be fully compliant within 5 μ s after tx_mode is set to DATA. During LPI mode, the common-mode shall be maintained to within ± 150 mV of the pre-LPI value.

72.7.1.5 Differential output return loss

For frequencies from 50 MHz to 7500 MHz, the differential return loss, in dB with f in MHz, of the transmitter shall meet the requirements of Equation (72–4) and Equation (72–5). This output impedance requirement applies to all valid output levels. The reference impedance for differential return loss measurements shall be 100 Ω .

$$ReturnLoss(f) \geq 9 \quad (72-4)$$

for 50 MHz $\leq f < 2500$ MHz

$$ReturnLoss(f) \geq 9 - 12 \log_{10} \left(\frac{f}{2500 \text{ MHz}} \right) \quad (72-5)$$

for 2500 MHz $\leq f \leq 7500$ MHz

The minimum differential output return loss is shown in Figure 72–9.

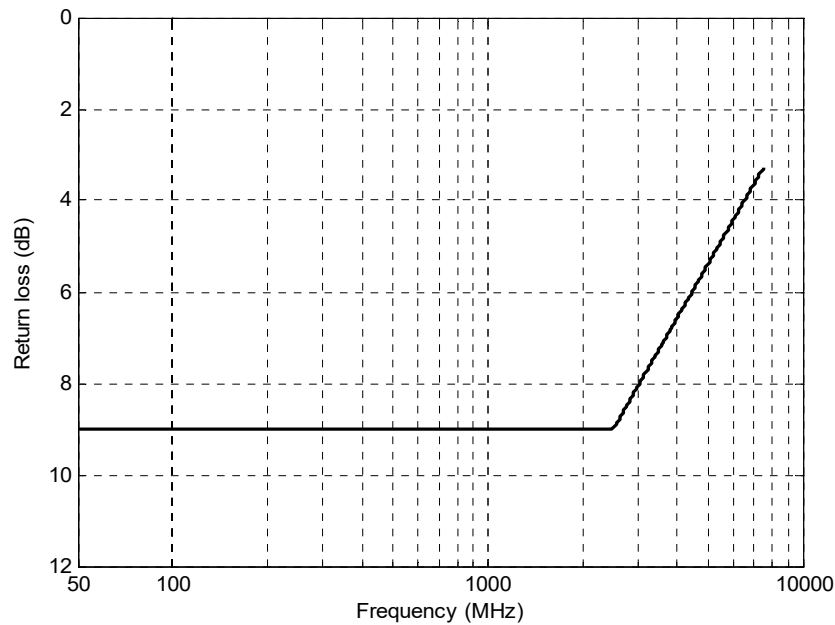


Figure 72–9—Minimum differential output return loss

72.7.1.6 Common-mode output return loss

The transmitter common-mode return loss shall meet the requirements of Equation (72–6) and Equation (72–7). The reference impedance for common-mode return loss measurements is 25 Ω.

$$ReturnLoss(f) \geq 6 \tag{72-6}$$

for 50 MHz ≤ *f* < 2500 MHz

$$ReturnLoss(f) \geq 6 - 12 \log_{10} \left(\frac{f}{2500 \text{ MHz}} \right) \tag{72-7}$$

for 2500 MHz ≤ *f* ≤ 7500 MHz

The minimum common-mode output return loss is shown in Figure 72–10.

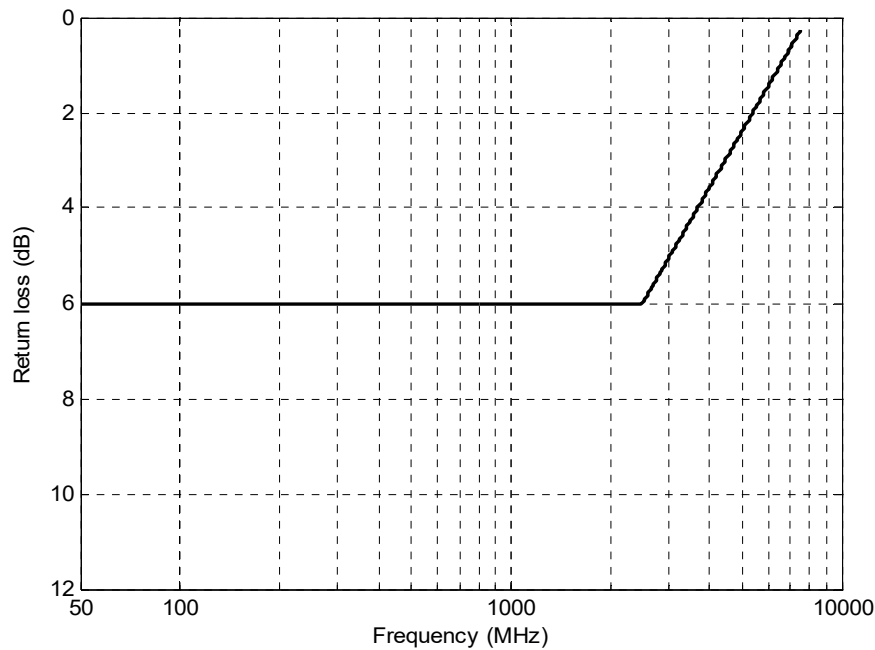


Figure 72–10—Minimum common-mode output return loss

72.7.1.7 Transition time

The rising and falling edge transition times shall be between 24 ps and 47 ps as measured at the 20% and 80% levels referenced to v_2 and v_5 as defined in 72.7.1.11. Measurement is done using the square wave test pattern defined in 52.9.1.2, with no equalization and a run of at least eight consecutive ones. Transmit equalization may be disabled by asserting the preset control defined in Table 45–77 and 45.2.1.97.3.

72.7.1.8 Transmit jitter test requirements

Transmit jitter is defined with respect to a test procedure resulting in a BER bathtub curve such as that described in Annex 48B.3. For the purpose of jitter measurement, the effect of a single-pole high-pass filter with a 3 dB point at 4 MHz is applied to the jitter. The data pattern for jitter measurements shall be test patterns 2 or 3 as defined in 52.9.1.1. Crossing times are defined with respect to the mid-point (0 V) of the AC-coupled differential signal. Equalization shall be off during jitter testing. Transmit equalization may be disabled by asserting the preset control defined in Table 45–77 and 45.2.1.97.3.

The duty cycle distortion test pattern shall consist of no fewer than eight symbols of alternating polarity (e.g., a 10101010... bit sequence).

NOTE—The required test patterns may be found in the training pattern field of the training frames or test patterns 2 or 3 as defined in 52.9.1.1.

72.7.1.9 Transmit jitter

The transmitter shall have a maximum total jitter of 0.28 UI peak-to-peak, composed of a maximum deterministic component of 0.15 UI peak-to-peak and a maximum random component of 0.15 UI peak-to-peak. Duty cycle distortion (DCD) is considered a component of deterministic jitter and shall not exceed 0.035 UI peak-to-peak. The peak-to-peak duty cycle distortion is defined as the absolute value of the difference in the mean pulse width of a 1 pulse or the mean pulse width of a 0 pulse (as measured at the mean of the high and

low voltage levels in a repeating 0101 bit sequence) and the nominal pulse width. Jitter specifications are specified for BER 10^{-12} . Transmit jitter test requirements are specified in 72.7.1.8.

72.7.1.10 Transmitter output waveform

The 10GBASE-KR transmitter includes programmable equalization to compensate for frequency-dependent loss in the backplane channel and facilitate data recovery at the receiver. This equalization may be accomplished with a three-tap finite impulse response (FIR) structure as shown in Figure 72–11. The actual implementation of the transmit equalizer, including the incorporation of additional taps, is beyond the scope of this standard.

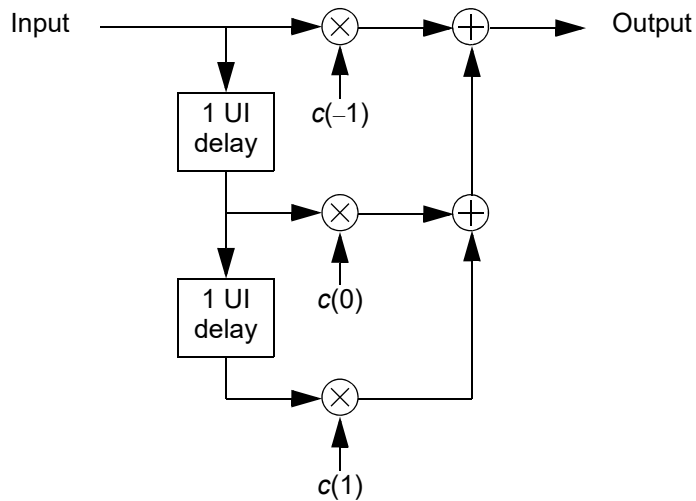


Figure 72–11—Transmit equalizer example

Transmit equalizer performance is specified in terms of the voltages defined in 72.7.1.11. It should be noted that the valid ranges of the $c(1)$ and $c(-1)$ coefficients may include positive and negative values. A value of zero is used to turn off equalization for the tap.

72.7.1.11 Transmitter output waveform requirements

The test pattern for the transmitter output waveform is the square wave test pattern defined in 52.9.1.2, with a run of at least eight consecutive ones. The transmitter output waveform test is based on the voltages v_1 through v_6 , Δv_2 , and Δv_5 , which shall be measured as shown in Figure 72–12 and described below.

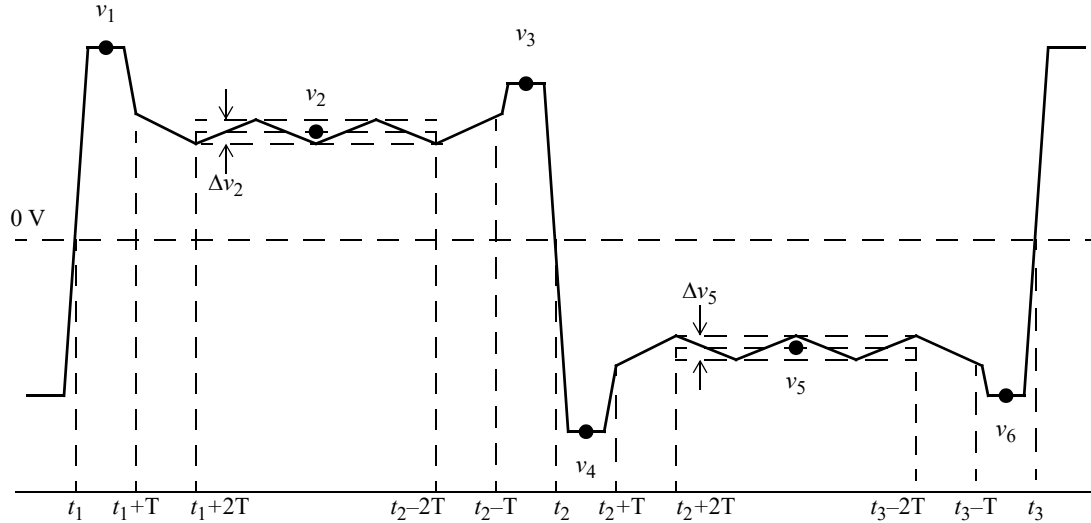


Figure 72–12—Transmitter output waveform

T	=	symbol period
t_1	=	zero-crossing point of the first rising edge of the AC-coupled signal
t_2	=	zero-crossing point of the falling edge of the AC-coupled signal
t_3	=	zero-crossing point of the second rising edge of the AC-coupled signal
v_1	=	maximum voltage measured in the interval t_1 to $t_1 + T$
v_2	=	positive steady-state voltage measured as the average voltage in the interval $t_1 + 2T$ to $t_2 - 2T$
v_3	=	maximum voltage measured in the interval $t_2 - T$ to t_2
v_4	=	minimum voltage measured in the interval t_2 to $t_2 + T$
v_5	=	negative steady-state voltage measured as the average voltage in the interval $t_2 + 2T$ to $t_3 - 2T$
v_6	=	minimum voltage measured in the interval $t_3 - T$ to t_3
Δv_2	=	positive voltage ripple measured as the peak-to-peak value of the difference between the voltage in the range $t_1 + 2T$ to $t_2 - 2T$ and v_2
Δv_5	=	negative voltage ripple measured as the peak-to-peak value of the difference between the voltage in the range $t_2 + 2T$ to $t_3 - 2T$ and v_5

From these voltages, the pre- and post-cursor equalization ratios R_{pre} and R_{pst} are derived from Equation (72–8) and Equation (72–9).

$$R_{\text{pre}} = \frac{v_3}{v_2} \quad (72-8)$$

$$R_{\text{pst}} = \frac{v_1}{v_2} \quad (72-9)$$

The state of the transmitter equalizer and hence the transmitter output waveform is manipulated via the protocol defined in 72.6.10 or via management. The changes in the transmitter output waveform resulting from coefficient update requests shall meet the requirements stated in Table 72–7. The coefficient update requests in Table 72–7 are to be followed by a coefficient update equal to hold for all taps. The results shall be verified after the coefficient status for all taps is reported as not_updated.

Table 72–7—Transmitter output waveform requirements related to coefficient update

Coefficient update ^a			Requirements ^b		
$c(1)$	$c(0)$	$c(-1)$	$v_1(k) - v_1(k-1)$ (mV)	$v_2(k) - v_2(k-1)$ (mV)	$v_3(k) - v_3(k-1)$ (mV)
increment	hold	hold	–20 to –5	5 to 20	5 to 20
decrement	hold	hold	5 to 20	–20 to –5	–20 to –5
hold	increment	hold	5 to 20	5 to 20	5 to 20
hold	decrement	hold	–20 to –5	–20 to –5	–20 to –5
hold	hold	increment	5 to 20	5 to 20	–20 to –5
hold	hold	decrement	–20 to –5	–20 to –5	5 to 20

^aStep size requirements for the tap under test apply regardless of the current value of the other taps.

^bThis difference is measured relative to the voltage prior to the assertion coefficient update k equal to hold.

For any coefficient update, the magnitudes of the changes in v_1 , v_2 , and v_3 shall be within 5 mV of each other. When sufficient increment or decrement updates have been applied to a given tap, it will reach a maximum or minimum limit governed by the coefficient range or by restrictions placed on minimum steady-state or maximum peak voltage, and the coefficient status is reported accordingly. The transmitter output waveform shall meet the requirements of Table 72–8 for all of the limiting cases represented in the table. Implementation of $c(-1)$ or $c(1)$ coefficient values greater than zero or less than the minimum defined by R_{pre} (min) and R_{pst} (min) is optional. A coefficient may be disabled by first asserting the preset control defined in Table 45–215 and 45.2.1.97, then manipulating the other coefficients as required by the test.

Table 72–8—Transmitter output waveform requirements related to coefficient status

Coefficient status			Requirements		
$c(1)$	$c(0)$	$c(-1)$	R_{pre}	R_{pst}	v_2 (mV)
disabled	minimum	disabled	0.90 to 1.10	0.90 to 1.10	220 to 330
disabled	maximum	disabled	0.95 to 1.05	0.95 to 1.05	400 to 600
minimum	minimum	disabled	—	4.00 (min)	—
disabled	minimum	minimum	1.54 (min)	—	—

In addition:

- The quantities Δv_2 and Δv_5 shall not exceed 40 mV peak-to-peak.

- b) The positive and negative voltages shall match such that each of the quantities $\min(|v_1 + v_4|/v_1, |v_1 + v_4|/|v_4|)$, $\min(|v_2 + v_5|/v_2, |v_2 + v_5|/|v_5|)$, and $\min(|v_3 + v_6|/v_3, |v_3 + v_6|/|v_6|)$ does not exceed 0.05.
- c) The quantity v_2 shall be greater than or equal to 40 mV.
- d) Any coefficient update equal to decrement applied to any tap that would result in v_2 less than 40 mV shall return a coefficient status value minimum.
- e) Any coefficient update equal to decrement that would result in a violation of 72.7.1.4 shall return a coefficient status value minimum for that coefficient.
- f) Any coefficient update equal to increment that would result in a violation of 72.7.1.4 shall return a coefficient status value maximum for that coefficient.

72.7.2 Receiver characteristics

Receiver characteristics at TP4 are summarized in Table 72–9 and detailed in 72.7.2.1 through 72.7.2.5.

Table 72–9—Receiver characteristics for 10GBASE-KR

Parameter	Subclause reference	Value	Units
Bit error ratio	72.7.2.1	10^{-12}	
Signaling speed	72.7.2.2	10.3125 ± 100 ppm	GBd
Receiver coupling	72.7.2.3	AC	
Differential input peak-to-peak amplitude (maximum)	72.7.2.4	1200 ^a	mV
Differential input return loss (minimum) ^b	72.7.2.5	[See Equation (72–4) and Equation (72–5)]	dB

^aThe receiver shall tolerate amplitudes up to 1600 mV without permanent damage.

^bRelative to 100 Ω differential.

72.7.2.1 Receiver interference tolerance

The receiver interference tolerance shall consist of two separate tests as described in Annex 69A with the parameters specified in Table 72–10. The data pattern for the interference tolerance test shall be the test patterns 2 or 3 as defined in 52.9.1.1. The receiver shall satisfy the requirements for interference tolerance specified in Annex 69A for both tests.

Table 72–10—10GBASE-KR interference tolerance parameters

Parameter	Test 1 values	Test 2 values	Units
Target BER	10^{-12}	10^{-12}	
m_{TC} (min.) ^a	1.0	0.5	
Amplitude of broadband noise (min. RMS)	5.2	12	mV
Applied transition time (20%–80%, min.)	47	47	ps
Applied Sinusoidal jitter (min. peak-to-peak)	0.115	0.115	UI
Applied random jitter (min. peak-to-peak) ^b	0.130	0.130	UI
Applied Duty Cycle Distortion (min. peak-to-peak)	0.035	0.035	UI

^a m_{TC} is defined in Equation (69A–6) of Annex 69A.

^bApplied random jitter is specified at a BER of 10^{-12} .

72.7.2.2 Signaling speed range

A 10GBASE-KR receiver shall comply with the requirements of Table 72–9 for any signaling speed in the range 10.3125 Gb/s $\pm$ 100 ppm.

72.7.2.3 AC-coupling

The 10GBASE-KR receiver shall be AC-coupled to the backplane to allow for maximum interoperability between various 10 Gb/s components. AC-coupling is considered to be part of the receiver for the purposes of this specification unless explicitly stated otherwise. It should be noted that there may be various methods for AC-coupling in actual implementations.

NOTE—It is recommended that the maximum value of the coupling capacitors be limited to 100 nF. This will limit the inrush currents to the receiver that could damage the receiver circuits when repeatedly connected to transmit modules with a higher voltage level.

72.7.2.4 Input signal amplitude

10GBASE-KR receivers shall accept differential input signal peak-to-peak amplitudes produced by compliant transmitters connected without attenuation to the receiver, and still meet the BER requirement specified in 72.7.2.1. Note that this may be larger than the 1200 mV differential maximum of 72.7.1.4 due to the actual transmitter output and receiver input impedances. The input impedance of a receiver can cause the minimum signal into a receiver to differ from that measured when the receiver is replaced with a 100 Ω test load. Since the channel is AC-coupled, the absolute voltage levels with respect to the receiver ground are dependent on the receiver implementation.

72.7.2.5 Differential input return loss

For frequencies from 100 MHz to 7500 MHz, the differential return loss, in dB with f in MHz, of the receiver shall be greater than or equal to Equation (72–4) and Equation (72–5). The reference impedance for differential return loss measurements is 100 Ω .

72.8 Interconnect characteristics

Informative interconnect characteristics for 10GBASE-KR are provided in Annex 69B.

72.9 Environmental specifications

72.9.1 General safety

Equipment subject to this clause shall conform to the general safety requirements in J.2.

72.9.2 Network safety

The designer is urged to consult the relevant local, national, and international safety regulations to ensure compliance with the appropriate requirements.

72.9.3 Installation and maintenance guidelines

It is recommended that sound installation practice, as defined by applicable local codes and regulations, be followed in every instance in which such practice is applicable.

72.9.4 Electromagnetic compatibility

A system integrating the 10GBASE-KR PHY shall comply with applicable local and national codes for the limitation of electromagnetic interference.

72.9.5 Temperature and humidity

A system integrating the 10GBASE-KR PHY is expected to operate over a reasonable range of environmental conditions related to temperature, humidity, and physical handling (such as shock and vibration). Specific requirements and values for these parameters are considered to be beyond the scope of this standard.

72.10 Protocol implementation conformance statement (PICS) proforma for Clause 72, Physical Medium Dependent (PMD) sublayer and baseband medium, type 10GBASE-KR¹³⁵

72.10.1 Introduction

The supplier of a protocol implementation that is claimed to conform to IEEE Std 802.3, Clause 72, Physical Medium Dependent (PMD) sublayer and baseband medium type 10GBASE-KR, shall complete the following protocol implementation conformance statement (PICS) proforma. A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

72.10.2 Identification

72.10.2.1 Implementation identification

Supplier	
Contact point for inquiries about the PICS	
Implementation Name(s) and Version(s)	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s)	
NOTE 1—Only the first three items are required for all implementations; other information may be completed as appropriate in meeting the requirements for the identification.	
NOTE 2—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

72.10.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3-2022, Clause 72, Physical Medium Dependent (PMD) sublayer and baseband medium type 10GBASE-KR
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3-2022.)	
Date of Statement	

¹³⁵Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

72.10.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
XGE	XGMII	72.1, 46	Interface is supported	O	Yes [] No []
XGXS	XGXS and XAUI	72.1, 47		O	Yes [] No []
PCS	Support of 10GBASE-R PCS	72.1, 49		M	Yes []
PMA	Support of 10 Gigabit serial PMA	72.1, 51		M	Yes []
AN	Auto-Negotiation for Backplane Ethernet	72.1, 73	Device implements Auto-Negotiation for Backplane Ethernet	M	Yes []
FEC	Forward error correction	72.1, 74	Device implements 10GBASE-R forward error correction	O	Yes [] No []
DC	Delay Constraints	72.4	Device conforms to delay constraints	M	Yes []
LPI	LPI	72.6.11	LPI	O	Yes [] No []
*MD	MDIO interface	72.5	Device implements MDIO	O	Yes [] No []
*TD	Global_PMD_transmit_disable	72.6.5		O	Yes [] No []

72.10.4 PICS proforma tables for Clause 72, Physical Medium Dependent (PMD) sublayer and baseband medium, type 10GBASE-KR

72.10.4.1 PCS requirements for AN service interface

Item	Feature	Subclause	Value/Comment	Status	Support
PR1	AN service interface primitive	72.3	The PCS associated with this PMD supports the AN service interface primitive AN_LINK.indication defined in 73.9	M	Yes []

72.10.4.2 PMD functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
FS1	Transmit function	72.6.2	Conveys bits from PMD service interface to MDI	M	Yes []
FS2	Transmitter signal	72.6.2	A positive differential voltage corresponds to tx_bit = ONE	M	Yes []
FS3	Receive function	72.6.3	Conveys bits from MDI to PMD service interface	M	Yes []
FS4	Receiver signal	72.6.3	A positive differential voltage corresponds to rx_bit = ONE	M	Yes []
FS5	Signal detect	72.6.4	Report to PMD service interface	M	Yes []
FS6	Global signal detect	72.6.4	Value described in 45.2.1.9.7	M	Yes []
FS7	SIGNAL_DETECT value	72.6.4	Set to FAIL	M	Yes []
FS8	SIGNAL_DETECT value	72.6.4	Set to OK when training is complete	M	Yes []
FS9	SIGNAL_DETECT value	72.6.4	Set to OK when training disabled	M	Yes []
FS10	Signal detect during LPI	72.6.4	Detect signal energy during LPI	LPI:M	Yes [] N/A []
FS11	Signal detect for EEE	72.6.4	Transition timing to set SIGNAL_DETECT	LPI:M	Yes [] N/A []
FS12	Transmit disable requirements	72.6.5	Requirements of 72.6.5 and Table 72–6	TD:M	Yes [] N/A []
FS13	Transmit disable during LPI	72.6.5	Disable transmitter during tx_mode = QUIET	LPI:M	Yes [] N/A []
FS14	Loopback support	72.6.6	Provided for 10GBASE-KR PMD by transmitter and receiver	M	Yes []

72.10.4.3 Management functions

Item	Feature	Subclause	Value/Comment	Status	Support
MF1	MDIO Variable Mapping	72.5	Per Table 72–2 and Table 72–3	MD:M	Yes [] N/A []
MF2	PMD_transmit_fault function	72.6.8	Sets PMD_transmit_fault as specified in 45.2.1.7.4	MD:M	Yes [] N/A []
MF3	PMD_receive_fault function	72.6.9	Sets PMD_receive_fault as specified in 45.2.1.7.5	MD:M	Yes [] N/A []

72.10.4.4 PMD Control functions

Item	Feature	Subclause	Value/Comment	Status	Support
CF1	Control Channel Encoding	72.6.10.2.2	Control channel transmitted using differential Manchester encoding (DME)	M	Yes []
CF2	Differential Manchester Encoding rules	72.6.10.2.2	Transitions at cell boundary.	M	Yes []
CF3	Differential Manchester Encoding rules	72.6.10.2.2	Presence of a mid-cell transition to signal logic 1	M	Yes []
CF4	Differential Manchester Encoding rules	72.6.10.2.2	Absence of a mid-cell transition to signal logic 0	M	Yes []
CF5	Coding violation	72.6.10.2.2	Ignore contents of control channel if coding violation found	M	Yes []
CF6	Coefficient update field format	72.6.10.2.3	Format of the coefficient update field per Table 72–4	M	Yes []
CF7	Cell 15 of the coefficient update field	72.6.10.2.3	Transmitted first	M	Yes []
CF8	Preset control	72.6.10.2.3.1	When received, pre-cursor and post-cursor coefficients set to zero	M	Yes []
CF9	Preset control	72.6.10.2.3.1	When received, main coefficient set to maximum value	M	Yes []
CF10	Preset control initially sent	72.6.10.2.3.1	Only when all coefficient status fields indicate not_updated and continues until all coefficients indicate updated or maximum	M	Yes []
CF11	Outgoing initialize control	72.6.10.2.3.1	Set to zero when all coefficients indicate updated or maximum following preset	M	Yes []
CF12	Maximum status	72.6.10.2.3.1	Returned when the main coefficient is updated	M	Yes []
CF13	Maximum status	72.6.10.2.3.1	Returned for pre-cursor and/or post-cursor coefficients when coefficient updated and zero is its maximum value	M	Yes []
CF14	Updated status	72.6.10.2.3.1	Returned for pre-cursor and/or post-cursor coefficients when the coefficient is updated and it supports additional settings above the value zero	M	Yes []
CF15	New Preset or Initialize requests	72.6.10.2.3.1	Not sent until the incoming status for all coefficients revert to not_updated	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
CF16	Preset	72.6.10.2.3.1	Not sent in combination with initialize or coefficient increment/decrement requests	M	Yes []
CF17	Initialize control	72.6.10.2.3.2	When received, taps set to meet conditions of 72.6.10.4.2	M	Yes []
CF18	Initialize control initially sent	72.6.10.2.3.2	Only when all coefficient status fields indicate not_updated and continues until all coefficients indicate updated	M	Yes []
CF19	Updated status	72.6.10.2.3.2	Returned for each coefficient when the coefficient update is complete	M	Yes []
CF20	Outgoing initialize field	72.6.10.2.3.2	Set to zero when all coefficients indicate update complete following initialize	M	Yes []
CF21	New Preset or Initialize requests	72.6.10.2.3.2	Not sent until the incoming status for all coefficients revert to not_updated	M	Yes []
CF22	Initialize	72.6.10.2.3.2	Not sent in combination with coefficient increment/decrement requests	M	Yes []
CF23	Increment or decrement encodings transmitted	72.6.10.2.3.3	Transmitted until status indicates: updated, maximum, or minimum	M	Yes []
CF24	Outgoing requests	72.6.10.2.3.3	Set to hold once update status for tap indicates updated, maximum or minimum	M	Yes []
CF25	Increment or decrement request	72.6.10.2.3.3	Not sent before status reverts to not_updated	M	Yes []
CF26	Encoding of coefficient update	72.6.10.2.3.3	Per Table 72–4	M	Yes []
CF27	Format of status report field	72.6.10.2.4	Per Table 72–5	M	
CF28	Cell 15 of the status report field	72.6.10.2.4	Transmitted first	M	
CF29	Receiver ready indication	72.6.10.2.4.4	Per Table 72–5	M	Yes []
CF30	Coefficient status	72.6.10.2.4.5	Per Table 72–5	M	Yes []
CF31	Training pattern length	72.6.10.2.6	512 octets	M	Yes []
CF32	Training pattern generator	72.6.10.2.6	Per Figure 72–3	M	Yes []
CF33	Training pattern seed	72.6.10.2.6	The pseudo-random generator shall have a random seed at the start of the training pattern	M	Yes []
CF34	Remote_rx_ready	72.6.10.3.1	TRUE after three or more consecutive training frames received with receiver ready indicated	M	Yes []
CF35	Wait Timer	72.6.10.3.2	100 to 300 training frames	M	Yes []
CF36	Max Wait Timer	72.6.10.3.2	500 ms $\pm$ 1%	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
CF37	Slip function to find framesync	72.6.10.3.2	Evaluates all possible positions	M	Yes []
CF38	Frame Lock state diagram	72.6.10.4.1	Meets requirements of Figure 72-4	M	Yes []
CF39	Training state diagram	72.6.10.4.2	Meets requirements of Figure 72-5	M	Yes []
CF40	Entry to INITIALIZE state	72.6.10.4.2	Transmitter equalizer configured per 72.6.10.4.2	M	Yes []
CF41	Initial value of $c(0)$ at the start of training	72.6.10.4.2	Meets the requirements of 72.6.10.4.2	M	Yes []
CF42	Coefficient Update state diagram	72.6.10.4.3	Meets requirements of Figure 72-6	M	Yes []

72.10.4.5 Transmitter electrical characteristics

Item	Feature	Subclause	Value/Comment	Status	Support
TC1	Test fixture impedance	72.7.1.2	100 Ω	M	Yes []
TC2	Differential return loss of test fixture	72.7.1.2	Per Equation (72-2) and Equation (72-3)	M	Yes []
TC3	Signaling speed	72.7.1.3	10.3125 GBd $\pm$ 100 ppm	M	Yes []
TC4	Maximum transmitter differential peak-to-peak voltage	72.7.1.4	Less than 1200 mV for a 1010 pattern	M	Yes []
TC5	Maximum transmitter differential peak-to-peak voltage when TX disabled	72.7.1.4	Less than 30 mV	M	Yes []
TC6	Common-mode output voltage	72.7.1.4	Between 0 and 1.9 V	M	Yes []
TC7	Output Amplitude LPI voltage	72.7.1.4	Less than 30 mV within 500 ns of tx_quiet	LPI:M	Yes [] N/A []
TC8	Output Amplitude ON voltage	72.7.1.4	Greater than 90% of previous level within 500 ns of tx_quiet de-asserted	LPI:M	Yes [] N/A []
TC9	Differential output return loss	72.7.1.5	Per Equation (72-4) and Equation (72-5)	M	Yes []
TC10	Differential output reference impedance	72.7.1.5	100 Ω	M	Yes []
TC11	Common-mode output return loss	72.7.1.6	Per Equation (72-6) and Equation (72-7)	M	Yes []
TC12	Rising edge transition time	72.7.1.7	Between 24 ps and 47 ps as measured at the 20% and 80% levels referenced to v_2 and v_5 as defined in 72.7.1.11 with no equalization	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
TC13	Falling edge transition time	72.7.1.7	Between 24 ps and 47 ps as measured at the 20% and 80% levels referenced to v_2 and v_5 as defined in 72.7.1.11 with no equalization	M	Yes []
TC14	Transmit jitter, peak-to-peak	72.7.1.9	See 72.7.1.8. Max TJ of 0.28 UI. Max DJ of 0.15 UI. Max RJ of 0.15 UI	M	Yes []
TC15	Duty Cycle Distortion	72.7.1.9	Not to exceed 0.035 UI	M	Yes []
TC16	Jitter test patterns	72.7.1.8	Test patterns 2 or 3 as defined in 52.9.1.1	M	Yes []
TC17	During jitter testing	72.7.1.8	Equalization turned off	M	Yes []
TC18	Changes in transmit output waveform resulting from coefficient updates	72.7.1.10	Meet requirements of Table 72–7	M	Yes []
TC19	Verification of coefficient updates	72.7.1.10	After the coefficient status for all taps is reported as not_updated	M	Yes []
TC20	Transmit output waveform	72.7.1.10	Meet requirements of Table 72–8	M	Yes []
TC21	v_2	72.7.1.10	Greater than or equal to 40 mV for all transmit equalizer configurations	M	Yes []
TC22	Coefficient status value minimum	72.7.1.10	Returned for any coefficient update equal to decrement applied to any tap that would result in Δv_2 or Δv_5 less than 40 mV	M	Yes []
TC23	Coefficient status value maximum	72.7.1.10	Returned for any coefficient update equal to decrement applied to c(-1) or c(1) that would result in a violation of 72.7.1.4	M	Yes []
TC24	Coefficient status value maximum	72.7.1.10	Returned for any coefficient update equal to increment applied to c(0) that would result in a violation of 72.7.1.4	M	Yes []
TC25	Transmitter output waveform	72.7.1.11	Verified with test patterns 2 or 3 as defined in 52.9.1.1	M	Yes []
TC26	$v_1, v_2, \Delta v_2, v_3, v_4, v_5, \Delta v_5, v_6$	72.7.1.11, 72.7.1.10	Measured per Figure 72–12. The absolute value of v_6 and v_3 are within 5%. The absolute value of v_1 and v_4 are within 5% and the absolute value of v_2 and v_5 are within 5%. The maximum peak-to-peak value of Δv_2 and Δv_5 shall not exceed 40 mV	M	Yes []

72.10.4.6 Receiver electrical characteristics

Item	Feature	Subclause	Value/Comment	Status	Support
RC1	Receiver amplitude tolerance	72.7.2	Amplitudes up to 1600 mV without permanent damage	M	Yes []
RC2	Receiver interference tolerance	72.7.2.1	Measured as described in Annex 69A with parameters in Table 72–10	M	Yes []
RC3	Receiver interference tolerance	72.7.2.1	Receiver interference tolerance test pattern per 72.7.2.1	M	Yes []
RC4	Receiver interference tolerance	72.7.2.1	Satisfy the requirements specified in Annex 69A	M	Yes []
RC5	Signaling speed	72.7.2.2	10.3125 GBd $\pm$ 100 ppm	M	Yes []
RC6	Receiver coupling	72.7.2.3	AC-coupled	M	Yes []
RC7	Input signal amplitude	72.7.2.4	BER still met when compliant transmitter is connected with no attenuation	M	Yes []
RC8	Differential return loss	72.7.2.5	Per Equation (72–4) and Equation (72–5)	M	Yes []

72.10.4.7 Environmental specifications

Item	Feature	Subclause	Value/Comment	Status	Support
ES1	General safety	72.9.1	Conforms to J.2	M	Yes []
ES2	Electromagnetic interference	72.9.4	Complies with applicable local and national codes	M	Yes []